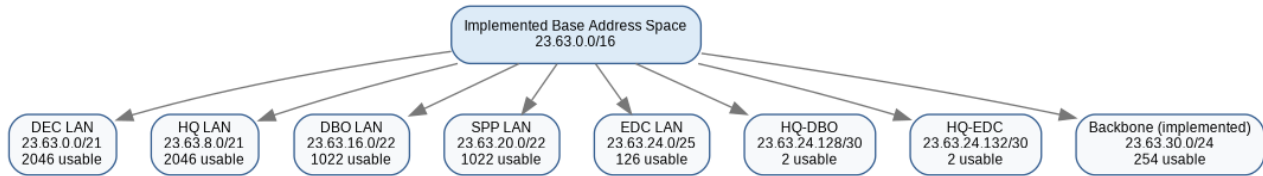


VLSM and Addressing Design

The implemented address space is based on 23.63.0.0/16. LAN subnet sizes were selected to meet the stated host requirements. Point-to-point links use /30 masks. The shared backbone is documented exactly as implemented and tested in Packet Tracer: 23.63.30.0/24.



Segment	Hosts Needed	Network/Prefix	Subnet Mask	Usable Hosts
DEC LAN	1200	23.63.0.0/21	255.255.248.0	2046
HQ LAN	2000	23.63.8.0/21	255.255.248.0	2046
DBO LAN	1000	23.63.16.0/22	255.255.252.0	1022
SPP LAN	700	23.63.20.0/22	255.255.252.0	1022
EDC LAN	120	23.63.24.0/25	255.255.255.128	126
HQ-DBO Link	2	23.63.24.128/30	255.255.255.252	2
HQ-EDC Link	2	23.63.24.132/30	255.255.255.252	2
Backbone	4	23.63.30.0/24	255.255.255.0	254
Segment	First Usable	Last Usable	Broadcast	
DEC LAN	23.63.0.1	23.63.7.254	23.63.7.255	
HQ LAN	23.63.8.1	23.63.15.254	23.63.15.255	
DBO LAN	23.63.16.1	23.63.19.254	23.63.19.255	
SPP LAN	23.63.20.1	23.63.23.254	23.63.23.255	
EDC LAN	23.63.24.1	23.63.24.126	23.63.24.127	
HQ-DBO Link	23.63.24.129	23.63.24.130	23.63.24.131	
HQ-EDC Link	23.63.24.133	23.63.24.134	23.63.24.135	
Backbone	23.63.30.1	23.63.30.254	23.63.30.255	

IP Addressing Tables

1.1 Router Interface Addressing

Router	Interface	IP Address	Prefix	Role
Router-HQ	G0/0	23.63.8.1	/21	HQ LAN
Router-HQ	G0/1	23.63.30.4	/24	Backbone
Router-HQ	S0/3/0	23.63.24.129	/30	HQ-DBO
Router-HQ	S0/3/1	23.63.24.133	/30	HQ-EDC
Router-DBO	G0/1	23.63.16.1	/22	DBO LAN
Router-DBO	G0/2	23.63.30.1	/24	Backbone
Router-DBO	S0/3/0	23.63.24.130	/30	HQ-DBO
Router-DEC	G0/0	23.63.30.2	/24	Backbone
Router-DEC	G0/1	23.63.0.1	/21	DEC LAN
Router-SPP	G0/0	23.63.30.3	/24	Backbone
Router-SPP	G0/1	23.63.20.1	/22	SPP LAN
Router-EDC	G0/0	23.63.24.1	/25	EDC LAN
Router-EDC	S0/3/0	23.63.24.134	/30	HQ-EDC

1.2 Server Addressing

Server	IP Address	Subnet Mask	Default Gateway	DNS
DNS Server	23.63.8.10	255.255.248.0	23.63.8.1	Local / self
Web Server	23.63.8.20	255.255.248.0	23.63.8.1	23.63.8.10
HQ Mail Server	23.63.8.30	255.255.248.0	23.63.8.1	23.63.8.10
SPP DHCP Server	23.63.20.2	255.255.252.0	23.63.20.1	23.63.8.10
SPP Mail Server	23.63.20.30	255.255.252.0	23.63.20.1	23.63.8.10

1.3 Verified DHCP Client Samples

Client	Leased IP	Mask	Gateway	DNS
PC DBO1	23.63.16.12	255.255.252.0	23.63.16.1	23.63.8.10
PC DEC1	23.63.0.11	255.255.248.0	23.63.0.1	23.63.8.10
PC SPP1	23.63.20.12	255.255.252.0	23.63.20.1	23.63.8.10

6. Routing Design and Implementation

6.1 RIPv2

All five routers run RIPv2 with no auto-summary. The common major-network statement is network 23.0.0.0. The route tables showed RIP-learned entries across the shared backbone and point-to-point links.

```
router rip
version 2
network 23.0.0.0
no auto-summary
```

6.2 Static Routing and DEC-to-HQ Policy

RouterDEC has a specific static route to the HQ LAN through RouterDBO. This overrides the direct RIP-learned backbone path to HQ and enforces the assignment policy that DEC reaches HQ only through DBO.

```
Router-DEC
ip route 23.63.8.0 255.255.248.0 23.63.30.1

Verified traceroute to 23.63.8.20:
1 23.63.30.1 (RouterDBO)
2 23.63.30.4 (RouterHQ backbone interface)
3 23.63.8.20 (Web Server)
```

6.3 HQ-DBO Redundancy

RouterDBO uses the direct serial route to HQ as the primary path and a higher-AD static route over the backbone as the backup. During testing, Serial0/3/0 was administratively shut down. The active route changed from AD 1 via 23.63.24.129 to AD 10 via 23.63.30.4, while the HQ Web Server remained reachable.

```
Primary: ip route 23.63.8.0 255.255.248.0 23.63.24.129
Backup: ip route 23.63.8.0 255.255.248.0 23.63.30.4 10
```

6.4 SPP Floating Static Route

The required floating-route administrative distance is calculated from the supplied group data:

```
AD = Number of courses of second-last member x Number of group members
    = 4 x 4
    = 16
```

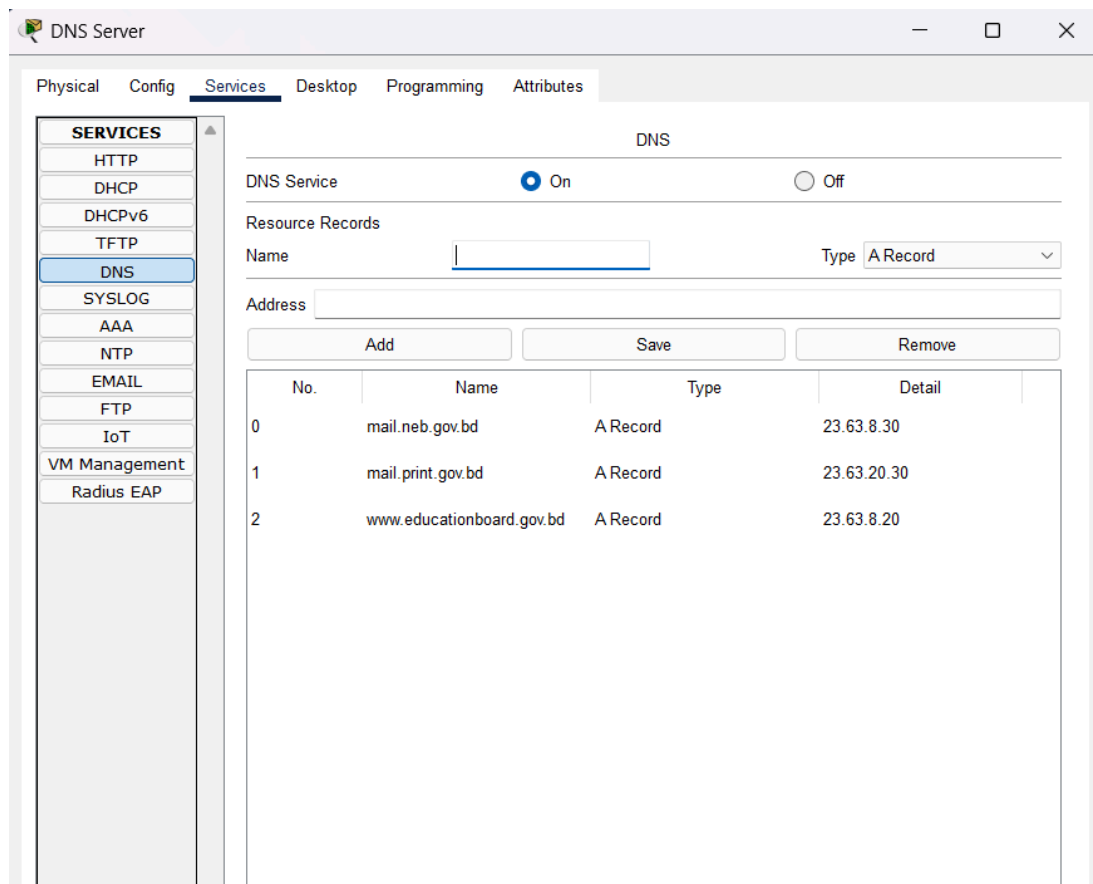
RouterSPP uses the direct HQ backbone next hop as the primary static route (AD 1) and RouterDBO as the floating backup (AD 16). The test temporarily removed the primary route; the AD-16 route became active and a 100% ping to 23.63.8.20 succeeded.

```
Primary: ip route 23.63.8.0 255.255.248.0 23.63.30.4
Backup: ip route 23.63.8.0 255.255.248.0 23.63.30.1 16
```

7. Server and Application Services

7.1 DNS

Name	Type	Address	Purpose
www.educationboard.gov.bd	A	23.63.8.20	Web Server
mail.neb.gov.bd	A	23.63.8.30	HQ Mail Server
mail.print.gov.bd	A	23.63.20.30	SPP Mail Server
educationboard.gov.bd	A	23.63.8.30	Mail-domain resolution
print.gov.bd	A	23.63.20.30	Mail-domain resolution



7.2 Web Service

HTTP/HTTPS was enabled on the HQ Web Server at 23.63.8.20. The required hostname is www.educationboard.gov.bd and the project page is configured for the message "National Examination Control System LIVE".

7.3 DHCP

DBO and DEC use router DHCP pools, while SPP uses the dedicated DHCP server. DNS option 23.63.8.10 is supplied to clients. Verified leases are shown in Section 5.3.

```
DBO_POOL: 23.63.16.0/22, gateway 23.63.16.1, DNS 23.63.8.10
DEC_POOL: 23.63.0.0/21, gateway 23.63.0.1, DNS 23.63.8.10
SPP_POOL: start 23.63.20.10, /22, gateway 23.63.20.1, DNS 23.63.8.10, max users 50
```

7.4 Email Service

Server	Domain	IP	Protocols	Accounts
HQ Mail Server	educationboard.gov.bd	23.63.8.30	SMTP ON; POP3 ON	hq1, hq2
SPP Mail Server	print.gov.bd	23.63.20.30	SMTP ON; POP3 ON	spp1, spp2

After correcting the SPP mail-domain setting to print.gov.bd and assigning DNS 23.63.8.10 to both mail servers, cross-site email communication was confirmed working between HQ and SPP.

```
Router-DEC#traceroute 23.63.8.20
Type escape sequence to abort.
Tracing the route to 23.63.8.20
```

1	23.63.30.1	0 msec	0 msec	!
2	23.63.30.4	8 msec	6 msec	!
3	23.63.8.20	6 msec	0 msec	!

```
Router-DEC#
```

DEC traceroute proving the path through RouterDBO before RouterHQ.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.educationboard.gov.bd
Ping request could not find host www.educationboard.gov.bd. Please check the name and try again.
C:\>
C:\>ping 23.63.8.10

Pinging 23.63.8.10 with 32 bytes of data:

Reply from 23.63.8.10: bytes=32 time=1ms TTL=125
Reply from 23.63.8.10: bytes=32 time=1ms TTL=125
Reply from 23.63.8.10: bytes=32 time=1ms TTL=125
Reply from 23.63.8.10: bytes=32 time=15ms TTL=125

Ping statistics for 23.63.8.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 15ms, Average = 4ms

C:\>ping www.educationboard.gov.bd

Pinging 23.63.8.20 with 32 bytes of data:

Reply from 23.63.8.20: bytes=32 time=1ms TTL=125
Reply from 23.63.8.20: bytes=32 time=15ms TTL=125
Reply from 23.63.8.20: bytes=32 time=15ms TTL=125
Reply from 23.63.8.20: bytes=32 time=10ms TTL=125

Ping statistics for 23.63.8.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 15ms, Average = 10ms

C:\>|
```

DNS reachability and hostname resolution to the Web Server.

8.1 DHCP Evidence

The screenshot shows a configuration window for PC DBO1. The 'Desktop' tab is selected, and the 'IP Configuration' section is highlighted. The interface is set to 'FastEthernet0'. Under 'IP Configuration', 'DHCP' is selected. The IPv4 settings are: IP Address 23.63.16.12, Subnet Mask 255.255.252.0, Default Gateway 23.63.16.1, and DNS Server 23.63.8.10. Under 'IPv6 Configuration', 'Static' is selected. The IPv6 settings are: IPv6 Address (empty), Link Local Address FE80::2D0:58FF:FE80:77DD, Default Gateway (empty), and DNS Server (empty). Under '802.1X', 'Use 802.1X Security' is unchecked. The Authentication is set to MD5, and the Username and Password fields are empty.

Section	Parameter	Value	
IP Configuration	Interface	FastEthernet0	
	Mode	DHCP	
	IPv4 Address	23.63.16.12	
	Subnet Mask	255.255.252.0	
IP Configuration	Default Gateway	23.63.16.1	
	DNS Server	23.63.8.10	
	IPv6 Configuration	Mode	Static
		IPv6 Address	
Link Local Address		FE80::2D0:58FF:FE80:77DD	
Default Gateway			
IPv6 Configuration	DNS Server		
	802.1X	Use 802.1X Security	Unchecked
Authentication		MD5	
Username			
Password			

DBO client DHCP lease.

PC DEC 1

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address: 23.63.0.11

Subnet Mask: 255.255.248.0

Default Gateway: 23.63.0.1

DNS Server: 23.63.8.10

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::260:47FF:FEBC:6589

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

DEC client DHCP lease.

. SPP client DHCP lease from the dedicated SPP DHCP service.

Router-HQ

8

```

!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
 ip address 23.63.8.1 255.255.248.0
 ip helper-address 23.63.20.10
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 23.63.30.4 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/3/0
 ip address 23.63.24.129 255.255.255.252
 clock rate 64000
!
interface Serial0/3/1
 ip address 23.63.24.133 255.255.255.252
 clock rate 64000
!
interface Vlan1
 no ip address
 shutdown
!
router rip
 version 2
 passive-interface GigabitEthernet0/0
 network 23.0.0.0
 no auto-summary
!
ip classless
ip route 23.63.16.0 255.255.252.0 23.63.24.130
ip route 23.63.0.0 255.255.248.0 23.63.24.130
ip route 23.63.20.0 255.255.252.0 23.63.24.130
ip route 23.63.24.0 255.255.255.128 23.63.24.134
ip route 23.63.16.0 255.255.252.0 23.63.30.1 10
!
ip flow-export version 9
!
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 login
!
!
!
end

```

Router-DBO

```

version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router-DBO
!
!
!
!
!
ip dhcp excluded-address 23.63.16.1 23.63.16.10
!

```

```

ip dhcp pool DBO_POOL
network 23.63.16.0 255.255.252.0
default-router 23.63.16.1
dns-server 23.63.8.10
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX1524QGZY-
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
shutdown
!
interface GigabitEthernet0/1
ip address 23.63.16.1 255.255.252.0
ip helper-address 23.63.20.10
duplex auto
speed auto
!
interface GigabitEthernet0/2
ip address 23.63.30.1 255.255.255.0
duplex auto
speed auto
!
interface Serial0/3/0
ip address 23.63.24.130 255.255.255.252
!
interface Serial0/3/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
router rip
version 2
network 23.0.0.0
no auto-summary
!
ip classless
ip route 23.63.8.0 255.255.248.0 23.63.24.129
ip route 23.63.24.0 255.255.255.128 23.63.24.129
ip route 23.63.8.0 255.255.248.0 23.63.30.4 10
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!

```

```
!  
end
```

Router-DEC

```
version 15.1  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname Router-DEC  
!  
!  
!  
!  
ip dhcp excluded-address 23.63.0.1 23.63.0.10  
!  
ip dhcp pool DEC_POOL  
network 23.63.0.0 255.255.248.0  
default-router 23.63.0.1  
dns-server 23.63.8.10  
!  
!  
!  
ip cef  
no ipv6 cef  
!  
!  
!  
license udi pid CISCO2911/K9 sn FTX1524GL33-  
!  
!  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
ip address 23.63.30.2 255.255.255.0  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/1  
ip address 23.63.0.1 255.255.248.0  
ip helper-address 23.63.20.10  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/2  
no ip address  
duplex auto  
speed auto  
shutdown  
!  
interface Vlan1  
no ip address  
shutdown  
!  
router rip  
version 2  
network 23.0.0.0  
no auto-summary  
!  
ip classless  
ip route 23.63.8.0 255.255.248.0 23.63.30.1  
!  
ip flow-export version 9  
!  
!  
!  
!  
!  
!  
line con 0
```

```

!
line aux 0
!
line vty 0 4
  login
!
!
!
end

```

Router-SPP

```

version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router-SPP
!
!
!
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX15243TGH-
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
  ip address 23.63.30.3 255.255.255.0
  duplex auto
  speed auto
!
interface GigabitEthernet0/1
  ip address 23.63.20.1 255.255.252.0
  ip helper-address 23.63.20.10
  duplex auto
  speed auto
!
interface GigabitEthernet0/2
  no ip address
  duplex auto
  speed auto
  shutdown
!
interface Vlan1
  no ip address
  shutdown
!
router rip
  version 2
  network 23.0.0.0
  no auto-summary
!
ip classless
ip route 23.63.16.0 255.255.252.0 23.63.30.2
ip route 23.63.24.0 255.255.255.128 23.63.30.2
ip route 23.63.8.0 255.255.248.0 23.63.30.1 16
ip route 23.63.8.0 255.255.248.0 23.63.30.4
!
ip flow-export version 9
!
!
!

```

```

!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
  login
!
!
!
end

```

Router-EDC

```

version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router-EDC
!
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX15247QSA-
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
  ip address 23.63.24.1 255.255.255.128
  ip helper-address 23.63.20.10
  duplex auto
  speed auto
!
interface GigabitEthernet0/1
  no ip address
  duplex auto
  speed auto
!
interface GigabitEthernet0/2
  no ip address
  duplex auto
  speed auto
!
interface Serial0/3/0
  ip address 23.63.24.134 255.255.255.252
!
interface Serial0/3/1
  no ip address
  clock rate 2000000
!
interface Vlan1
  no ip address
!
router rip
  version 2
  network 23.0.0.0
  no auto-summary
!

```

```

ip classless
ip route 23.63.8.0 255.255.248.0 23.63.24.133
ip route 23.63.16.0 255.255.252.0 23.63.24.133
ip route 23.63.0.0 255.255.248.0 23.63.24.133
ip route 23.63.20.0 255.255.252.0 23.63.24.133
!
ip flow-export version 9
!
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
  login
!
!
!
end

```

Assumptions Made

The following assumptions were made during the design and implementation of the Bangladesh Board Examination Emergency Network (BBEEN):

1. The required device numbers for HQ, DBO, DEC, SPP, and EDC were considered as host-capacity requirements for subnet design, rather than physically placing thousands of devices in Packet Tracer.
2. Only a limited number of representative PCs, printers, servers, switches, and routers were used in the Packet Tracer topology to demonstrate the complete network functionality.
3. The network was designed using the base addressing scheme 23.63.0.0/16, and VLSM was applied according to the host requirements of the different sites.
4. The first usable IP address of each LAN subnet was generally assigned to the corresponding router interface as the default gateway.
5. Servers were assigned static IP addresses so that DNS, Web, Mail, and DHCP services would remain consistently reachable.
6. Client PCs at DEC and DBO were assumed to obtain IP configuration through router-based DHCP, while SPP clients were assumed to obtain their addresses from the dedicated SPP DHCP server.
7. The common backbone network was implemented as 23.63.30.0/24 in the final Packet Tracer topology and documentation because this is the subnet used in the tested implementation.
8. The HQ–DBO and HQ–EDC point-to-point links were assigned /30 networks because each link requires only two usable router addresses.
9. The HQ was assumed to host the centralized DNS, Web, and Education Board Mail services, while the SPP hosted its dedicated DHCP and Mail services.
10. DNS resolution was assumed to be centralized through the DNS server at 23.63.8.10.
11. The Web Server was assumed to be accessed using www.educationboard.gov.bd, while the two mail systems use mail.neb.gov.bd and mail.print.gov.bd.
12. RIPv2 was used for dynamic route exchange where required, while static and floating static routes were used where explicitly required by the project specification.
13. DEC was configured so that traffic destined for the HQ network passes through DBO, satisfying the required DEC → DBO → HQ communication path.
14. The floating static route Administrative Distance was assumed/calculated as:

AD = Number of courses of second-last group member × Number of group members = 4 × 4 = 16

15. No default routes were used in the network, and all required destinations were reached through connected, static, floating static, or RIPv2 routes.
16. Unused router interfaces were kept shut down, and switches were not assigned management IP addresses because switch remote-management configuration was not required.
17. Successful ping, traceroute, DNS, HTTP, DHCP, email, primary-route, and backup-route tests were considered sufficient evidence of end-to-end connectivity and service availability.